BANF

Intelligent Tire System













shyou@banf.ai

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BANF: Intelligent Tire System

Real-time tire sensing that connects AV/EVs, fleets, OEMs, and tire manufacturers with actionable tire insights

- BANF delivers an intelligent tire system that connect AV/EVs, fleets, OEMs, and tire manufacturers
- By using real-time tire sensing technology, BANF provides tire intelligence that enhances safety, performance, efficiency, and maintenance for next generation mobility

Recent PR on BANF's partnerships and collaborations

KORE — View Article

VW NIRA DYNAMICS — View Article

HYUNDAI MOTOR GROUP — View Article

CampX BY VOLVO GROUP — View Article

DHL — View Article

CES Innovation Awards 2026 Honoree / CES INNOVATION AWARDS 2024 HONOREE — View Article

TC — View Article

Tire-Related Accidents: Significant Portion of Motor Vehicle Fatalities

Main Causes of Traffic Accidents

1st : Driver Error

2nd : Tire-related

3rd : Road

※ Source: Crosley Law, Tire Defects Cause Fatal Car Accidents Nationwide, 2023

Tire-related incidents encompass various factors such as **tire blowouts, tread separation, and improper tire maintenance**, all of which can lead to hazardous situations on the road. Addressing tire-related issues through proactive maintenance and adherence to safety protocols is crucial for mitigating the risk of accidents and ensuring safer roadways.

※ Source: NHTSA, Tire-Related Factors in the Pre-Crash Phase, 2023

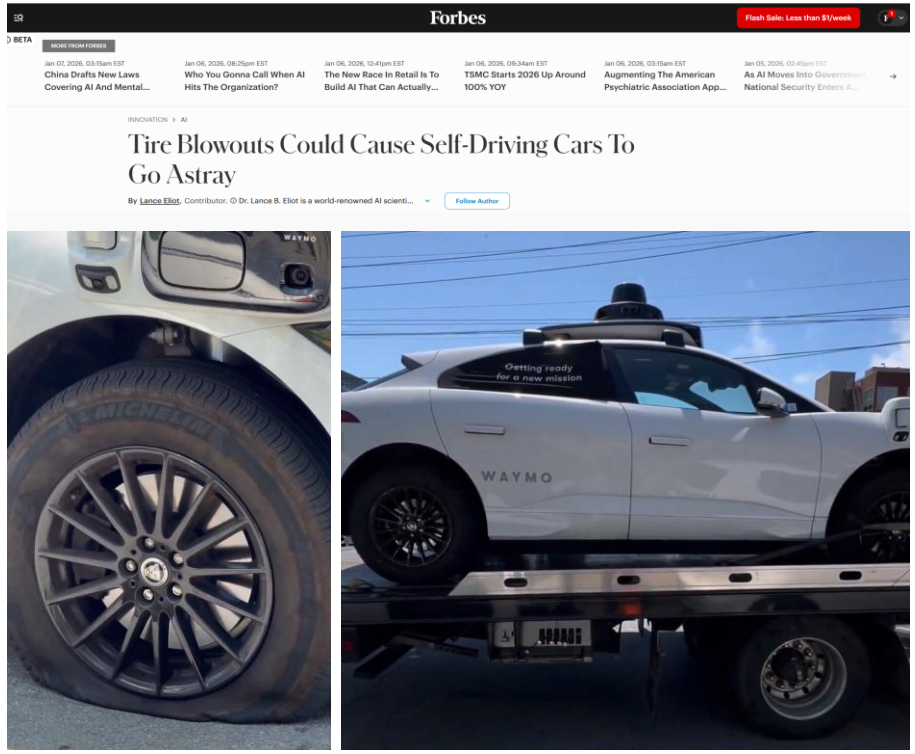


Autonomous Vehicles Tire-Related Incidents

Tire detachment incidents in AVs → Growing importance of tire monitoring in AV operations

- Real-world case of tire and wheel detachment during autonomous vehicle operation
- Long operating hours of AVs and robotaxis significantly increase tire-related risk

→ [Robotaxi · Autonomous Fleets: Intelligent tire monitoring systems are essential]

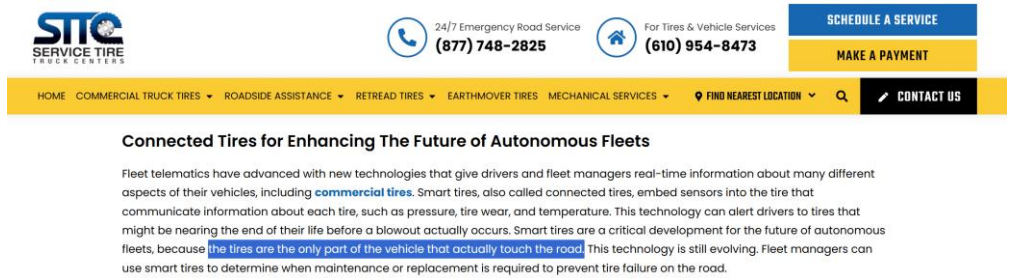


KAI WILLIAMS
SEP 18, 2025

97 41 12 Share

Everything was fine until the wheel came off.

At 1:14 AM on May 31st, a Waymo was driving on South Lamar Boulevard in Austin, Texas, when the front left wheel detached. The bottom of the car scraped against the pavement as the car skidded to a stop, and the passenger suffered a minor injury, according to Waymo.



Future Automotive Market Change

EV & AV expansion → Increasing tire load and stress



EV (Electric Vehicles)

- Heavier weight (large battery) → Increased tire load
- Strong acceleration (higher Torque) → Greater wear during acceleration

AV (Autonomous Vehicles)

- 24/7 Operation → Continuous tire stress and heat
  → Increased exposure to road impacts

Increasing Tire Load & Stress

With the spread of EV and AV, real-time monitoring of tire condition and road environment is becoming essential

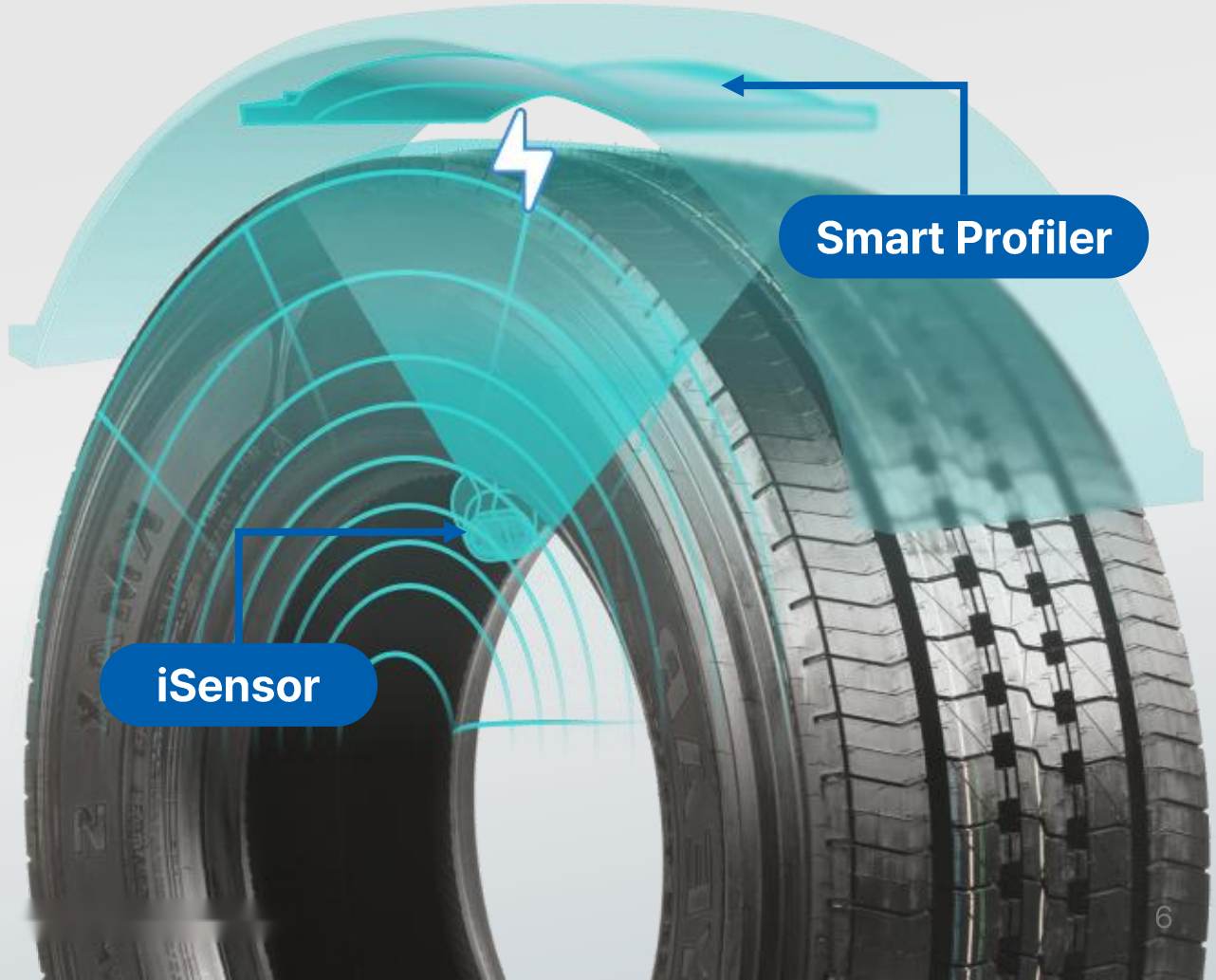


Real-Time Tire Sensing Technology with Tire Physical AI

Using iSensor and Smart Profiler, tire and road condition monitoring system

Real-Time Tire Safety Management with Tire AI

- Tire Pressure (Burst Detection: 40ms)
- Tire Temperature (Overheating)
- Wheel Alignment (Fuel Saving)
- Road surface
- Tire Load Measurement (Load Balance)
- Tire Tread Measurement



Limits of Existing Players in commercializing

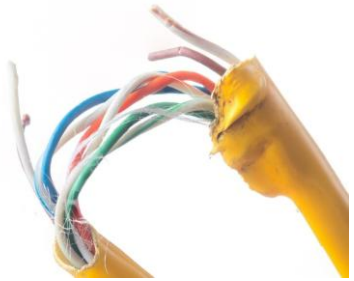
Stable Power Supply to Sensor: Prerequisite for Intelligent Tire Systems

- Continuous, reliable power is required to keep collecting high-resolution sensing data
- Sufficient power should be supplied to the sensor for real-time AI analysis and wireless communication
- A power solution that covers the entire tire lifecycle is needed to enable long-term operation without battery replacement

Power Supply Attempts

Many power supply concepts tried, but **none have commercialized successfully**

Wired Power Supply



Wire Breakage

Inductive Method



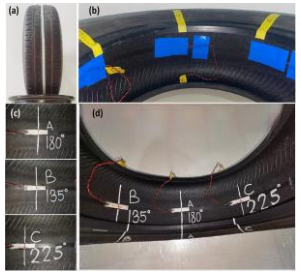
Max. 40mm Distance

RF



Max. 10uW Power

Energy Harvesting



Inoperative Below 30 km/h

Coin Battery



Limited Capacity

BANF Intelligent Tire System (20+ IPs)

Wireless sensor charging using magnetic resonance and real-time tire data collection

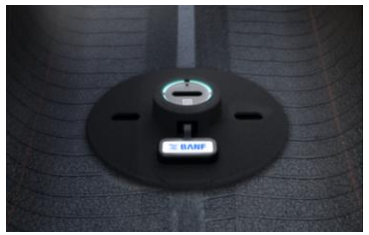
<Intelligent Tire System Architecture>

Smart Profiler



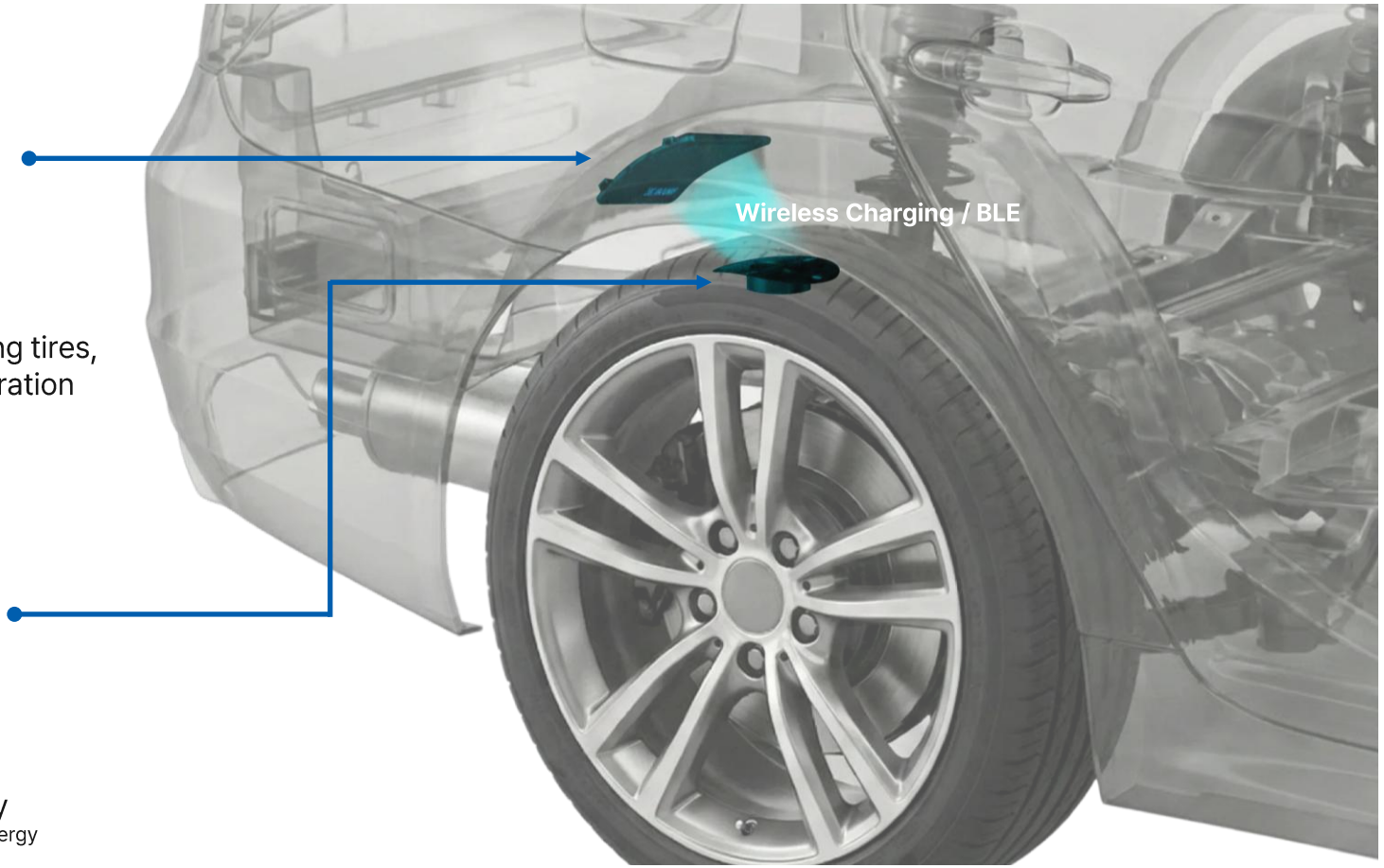
Providing wireless power to sensors inside fast-spinning tires,
directly linked to the vehicle power for seamless operation

iSensor



A real-time detection sensor featuring
3-axis acceleration sensor, TPMS, and BLE connectivity

※ BLE: Bluetooth Low Energy



iSensor: Accelerometer data collection

High-Speed Sampling, Precise Measurement of 3-Axis (X, Y, Z) Tire acceleration data

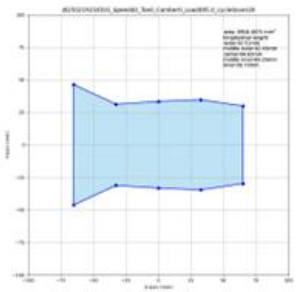
iSensor

- **Measurable acceleration**: 400g+
- **Sampling rate of the iSensor**: 1kHz – 4kHz
- **Measured Data**: Tire Pressure & Temperature, 3-axis acceleration
- **Wireless Communication**: BLE (Bluetooth Low Energy)

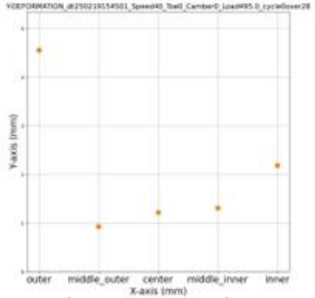


Quantitative Tire Performance Analysis

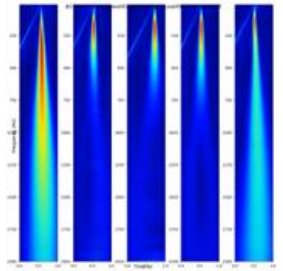
- Using 3-axis iSensor data, analyze **quantitative tire performance**
- **Analysis list**: Contact patch shape, tread behavior proxy, cornering stability proxy, rolling resistance proxy, vibration-to-noise proxy
- These metrics support **tire benchmarking, design optimization, and etc.**



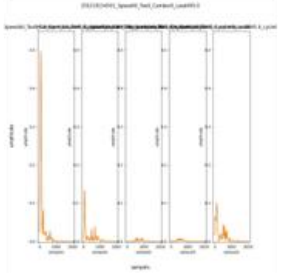
Contact Patch Shape



Tire Deformation



CWT



PSD

Standard TPMS vs. BANF's Next-Gen TPMS

BANF System: Next-Generation Performance in update rate, accuracy, and AI capabilities

Feature	Standard TPMS	BANF Intelligent Tire System	Comparison
Data Update Frequency	1-2 minutes	**20 milliseconds (50 Hz)**	3000-6000x
Pressure Change Detection	±1 PSI	**±0.1 PSI**	10x
Response Time to Blowout	60-120 seconds	**20-40 milliseconds**	~3000x
Slow Leak Detection	60 minutes	**Seconds**	~120x
Dual Wheeler Auto-Location	Limited or Manual	**Automatic via smart profiler**	Unique Solution
AI Features	X	**Load, Toe Angle, Road, etc.**	Unique Solution

Business Impact of BANF Intelligent Tire System

Intelligent tire system leveraging tire and road data to improve fuel efficiency, tire life, and safety

Economic Impact



Tire Pressure

- 1% of under-Inflation = **+ 0.3% Fuel consumption**
- Proper inflated tire = **Save $1,000 / Year**

※ Source: Tyrepower, How does tyre pressure influence your fuel economy, 2024



Wheel Alignment

- Misaligned wheels
  - Reduce fuel efficiency by **10%** and **shorten tire life**
- Correct alignment = **Save $2,000 / Year**

※ Source: GMI, Connected Tire Market, 2023



Load

- Balanced loads
  - Reduce tire stress, extend tire life, and cut breakdown
- **Save up to $1,500 per year**
  - On tire-related maintenance and roadside events

※ Source: Pedigree, Boost Fleet Savings & Safety with Tire sensor, 2025

Safety Impact



Temperature (Overheating)



Road Surface (Slippery Road, Pothole Depth)



Tread Depth



Loosen Wheel

BANF Global Recognition & Awards

Validated by leading global organizations – CES, Forbes, US Army, TechCrunch, and SAE International

CES Innovation Awards

- BANF TireSafe: Predictive Blowout and Leak Monitoring (2026)
- BANF iSensor: Real-time tire profile for fleets and AVs (2024)

Forbes Ais "100 to Watch"

- Selected to Forbes Asia's 2025 "100 to Watch" list as a high-potential industrial technology startup

U.S. Army xTech Live at CES 2025

- Winner of the US Army's xTech Live Pitch competition for intelligent tire technology for autonomous and connected military vehicles

TechCrunch Disrupt Startup Battlefield Top 20

- Selected Top 20 as Korea's first company in the 2024 TechCrunch Disrupt Battlefield

SAE International - GAMIC Global Automotive & Mobility Innovation Challenge

- Recognized as Korea's first innovative mobility startup in SAE's Global Automotive & Mobility Innovation Challenge, advancing intelligent tire technology

Global Awards and Honors Lists

Organization	Award Title	Award Details
CTA	**CES 2026**	**Innovation Award**
Forbes	**Asia 100 Companies to Watch**	
US Army	**US Army xTech Live**	**Winner**
TechCrunch	**Battlefield Top 200**	**Top 20**
The Circle FC	Delhi Demoday	우승
DHL	Fastforward Challenge	Top 3
SAE International	**GAMIC**	**Top 10**
Hannover Messe	Innovation World Cup	Top 18
MWC	4YFN Startup Award	Top 50
CTA	CES Innovation Award	Vehicle Tech
Hannover Messe	Innovation World Cup	Top 18
IRF	GRAA Startup Award	Grand Prize
ITRI	Mobileheroes 2023	Grand Prize
Hangzhou Municipal Government	Hangzhou I&E	Top 3
Alibaba Cloud	Create@ 2023	Innovation Award

Key Achievements

Partnerships

- Vehicle Manufacturer : 8
- Tire Company : 7
- Fleet Operator : 15



2021
- Raised Seed

2022
- Revenue: $100,000

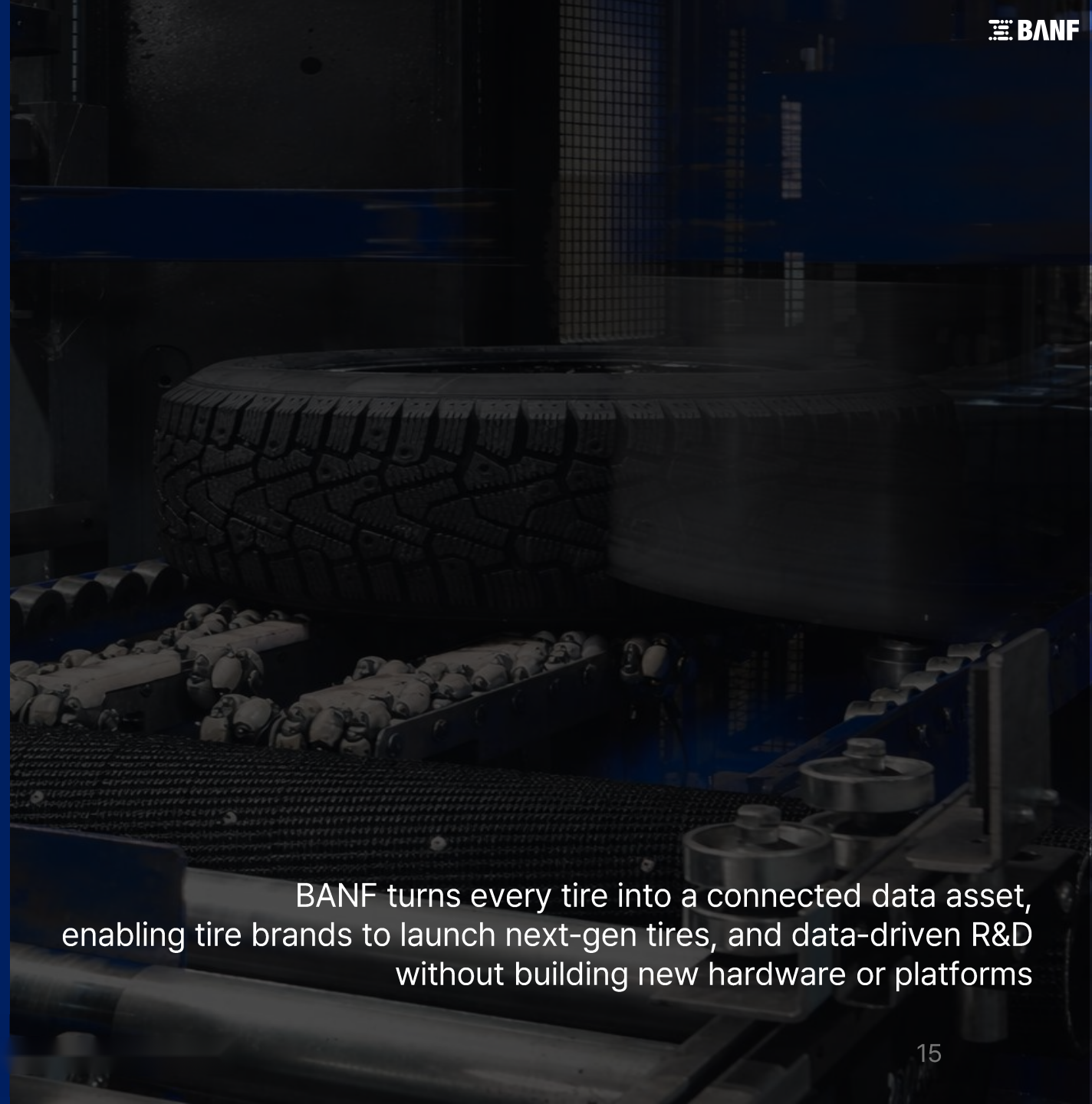
2023
- Revenue: $550,000
- Raised Pre-A ($5M)
- Advanced Pilot Launch

2024
- Revenue: $2,200,000
- CES Innovation Award
- PPAP Preparation

2025
- Revenue: $6,000,000
- Forbes 100 to Watch
- Pre-Production

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Tire Manufacturer

- Co-Development Next-Generation Tires
- Co-Development Sensor-Embedded Premium Tires



BANF turns every tire into a connected data asset,
enabling tire brands to launch next-gen tires, and data-driven R&D
without building new hardware or platforms

Co-Developing Next-Generation Tires

From tire benchmarking to Data-driven co-development with BANF

Phase 1 – Benchmarking Test

MICHELIN Continental BRIDGESTONE NEXEN TIRE GOODYEAR HANKOOK driving emotion

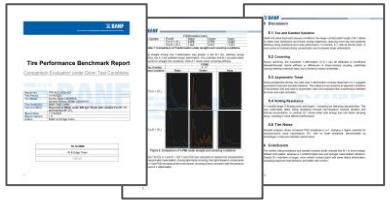
VS



Competitor
In-house Tire Line-up

Reference Tire

BANF Tire Performance Analysis Report
(Quantitative)



- Quantitatively benchmark Competitor or inhouse tires against the reference tire using BANF's Tire Performance analysis
- Evaluate key metrics under realistic load, speed, and road conditions
- Derive insights to optimize next-generation tire design, pattern, and compound

Phase 2 – Data-Driven Tire Co-Development



Reference Tire Model

+



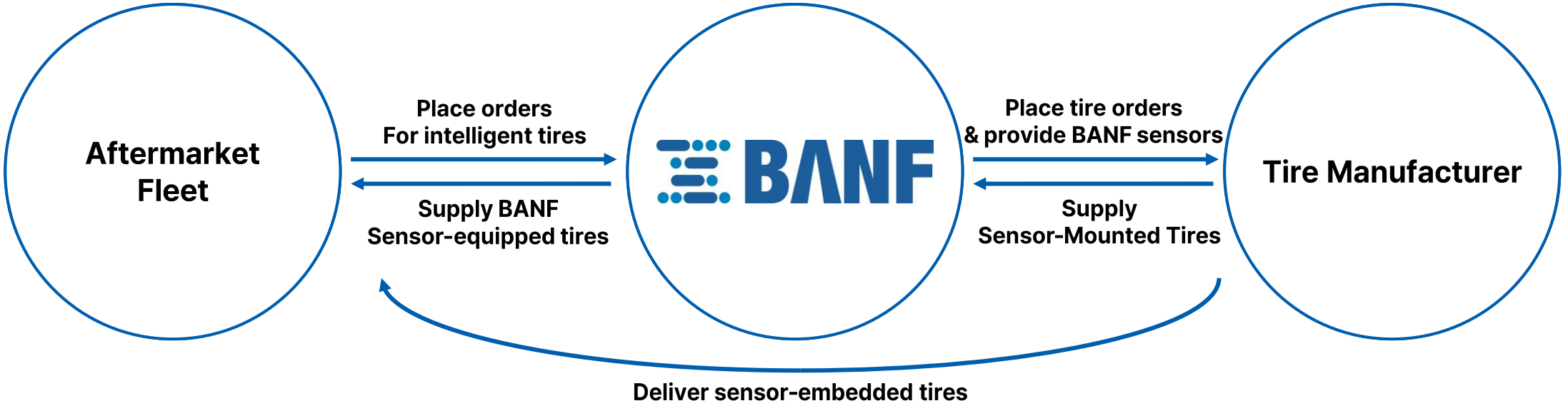
Tire Performance Benchmark Data
Tire Sensing Data

Next-Gen Optimized Tire Model (Co-Developed)

- Quantitatively refine the reference tire model using BANF's tire performance benchmark data
- Integrate real-world tire sensing and road-condition insights into tire design and simulation models
- Co-develop a next-gen optimized tire model that guides future patterns, compounds, and vehicle-specific tuning

Co-Developing Sensor-Embedded Premium Tires

Intelligent premium tires measuring pressure, temperature, wear, load, and road conditions



- BANF and the tire manufacturer co-develop sensor-embedded premium tire models
- The tire manufacturer produces and supplies sensor-mounted premium tires to fleets
- BANF provides solution services and shares incremental premium tire revenue with the manufacturer

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Autonomous Vehicle

- Tire Data-Driven Chassis Control for AVs



Real-time tire and road data
that enhance AV chassis, braking, and stability control
More precise and predictable

Tire Data-Driven Chassis Control for AVs

Using tire data to optimize braking, steering, and stability for autonomous vehicles

Safety Requirements for AVs

- Tire safety information → Real-time TPMS
- Tire data for advanced motion control → Road and Load classification

How to use BANF tire data

Route planning

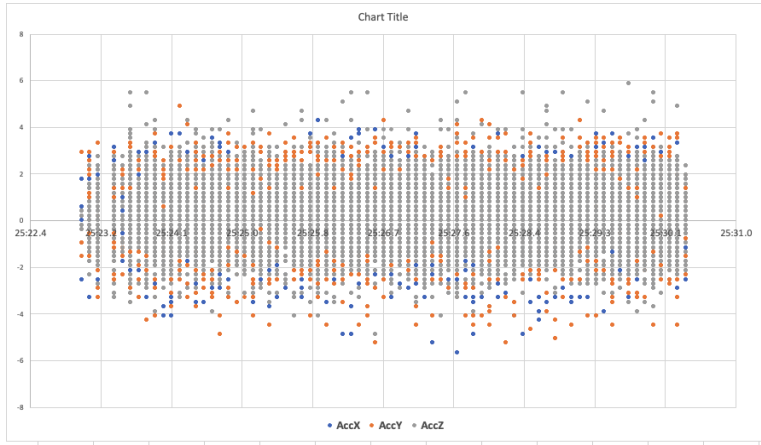
- Use real-time tire friction and load data to optimize safer on low-μ roads and in harsh weather conditions

Chassis Control

- Continuously adapt braking, steering, and stability control based on the estimated tire-road conditions

Redundancy & safety

- Add a sensing layer that complements cameras, lidars, and radars, enhancing AV safety and robustness in edge cases



Real-time collection of tire and road surface data (friction, wear, load, slip angle, etc.)



Real-time integration into autonomous driving chassis control algorithms (braking, steering, and stability control)

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Vehicle OEM

- Intelligent Tire System for SDV & ADAS

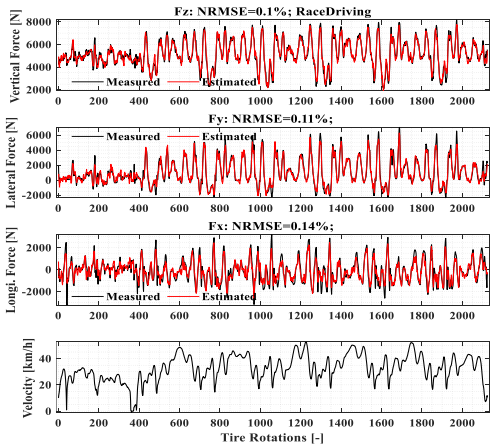


Intelligent tire system for SDV & ADAS
that boosts safety, range, and ride comfort

Intelligent Tire System for SDV & ADAS

Turning tire and road data into smarter vehicle control

BANF Intelligent Tire System Signal



- F_z(Vertical force)
- F_y(Lateral force)
- F_x(Longitudinal force)
- Velocity

Integration Concept

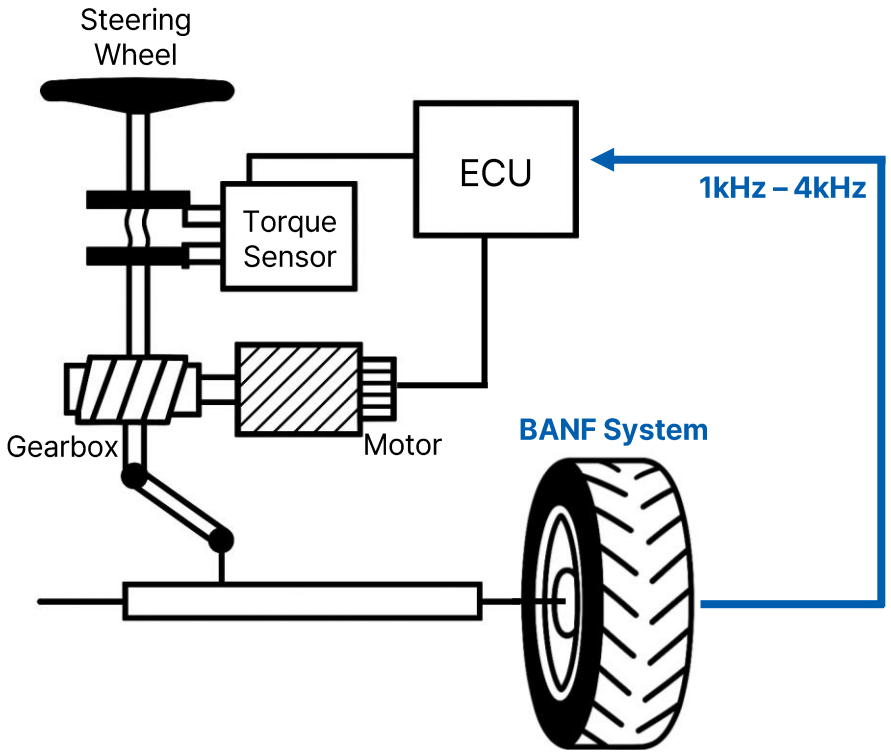
BANF Sensor Information	SDV/ADAS Integration Purpose
Tire Temperature & Pressure	Upgrade basic TPMS into high-accuracy tire health monitoring for SDV/ADAS
Road Surface (8 Different Conditions)	Provide real-time road friction and surface classification for ACC, AEB, and AV planning
Tire Tread Measurement	Enable tread-wear-based warnings and maintenance/tire-replacement services
Tire Load Management	Adapt suspension and stability control to real-time payload, improving safety and comfort
Wheel Alignment	Detect alignment issues and under/oversteer tendencies to keep AV/ADAS models accurate

- BANF intelligent tire system adds a dedicated tire-and-road intelligence layer to the vehicle platform
- SDV and ADAS functions gain access to friction, load, and wear information at the tire–road contact patch
- OEMs can enhance vehicle dynamics and safety, and unlock new connected-maintenance and fleet service offerings

Smart Steering with BANF Tire Data

Example scenario of how SDV/ADAS adapts steering using tire and road condition data

Steering (EPS) + BANF System Process



1. Detection & Tire Data Transmission

- As the driver turns the wheel on a wet or icy road, the BANF system instantly senses tire forces and road friction
- This tire and road information is sent in real time to the steering control ECU

2. Data Processing & Steering Control

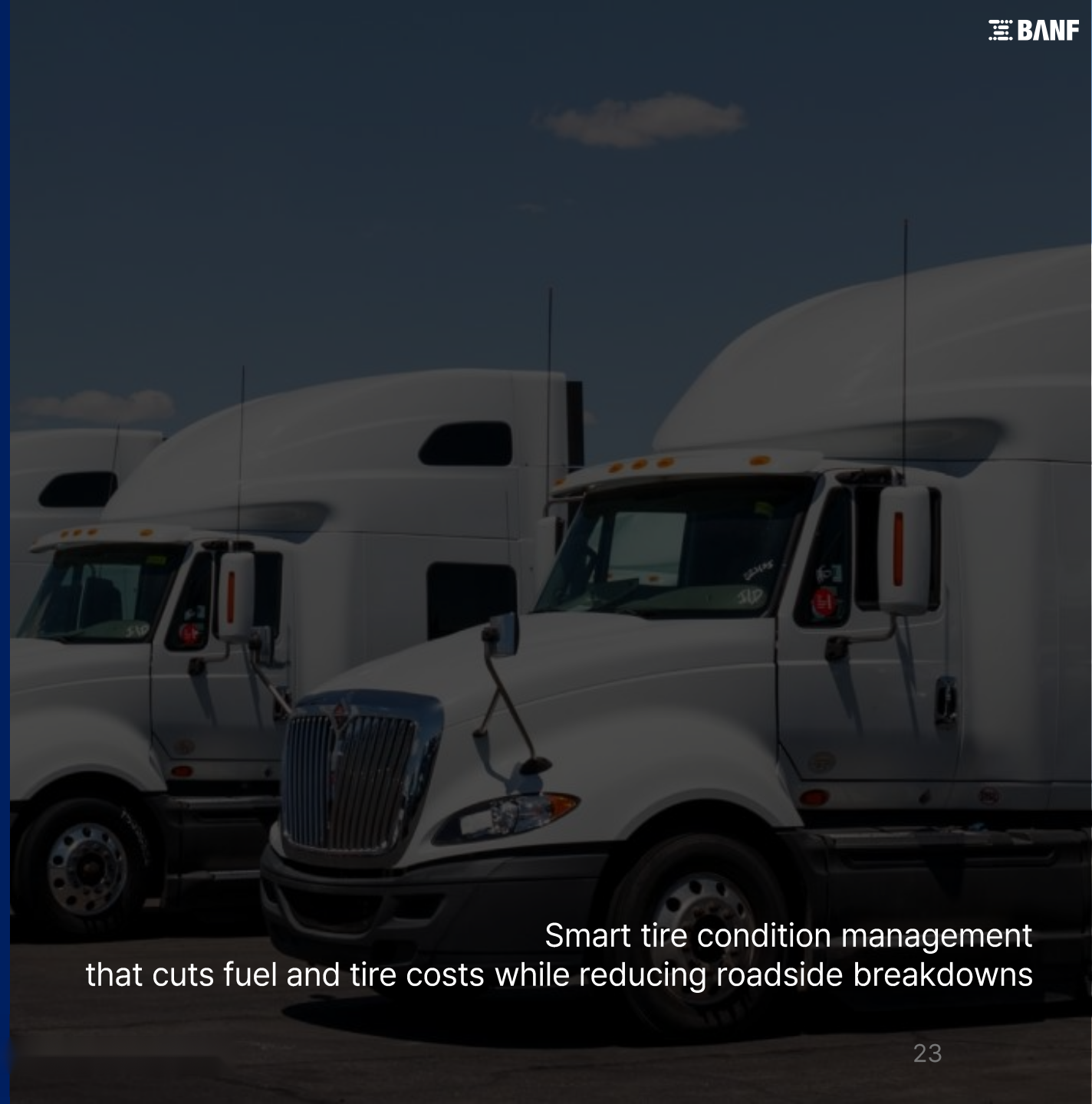
- The ECU combines BANF tire data with steering torque to understand how much grip is actually available
- If friction is low or load is high, it adjusts assist torque, road-feel, and steering damping to keep the vehicle stable and predictable

3. Steering & Vehicle Response

- The driver feels a more confident, predictable steering response instead of sudden over/under steer
- For ADAS/AV functions, the vehicle can keep its lane and follow its path more safely even on low-friction roads

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Fleet Operator

- Smart Fleet & Tire Management with BANF



Smart tire condition management
that cuts fuel and tire costs while reducing roadside breakdowns

Smart Fleet & Tire Management with BANF

Install once and monitor tire condition through FMS, moving from alerts to predictive tire maintenance

Fleet Application Architecture

1. On-Vehicle Installation

- Install BANF system on vehicles to monitor tire condition in real time

2. FMS Integration

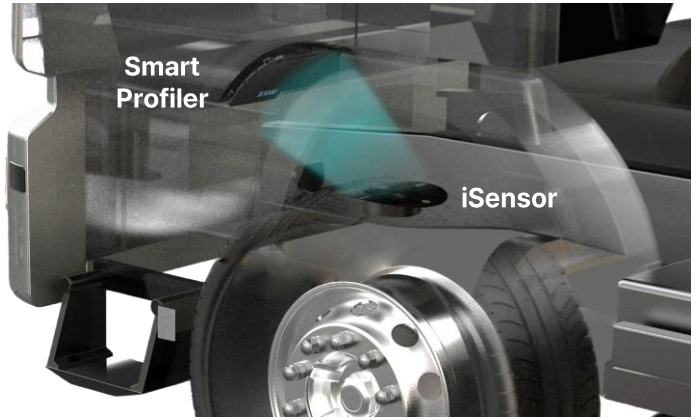
- Stream tire condition into BANFleet or your existing FMS

3. Early Condition Detection & Alerts

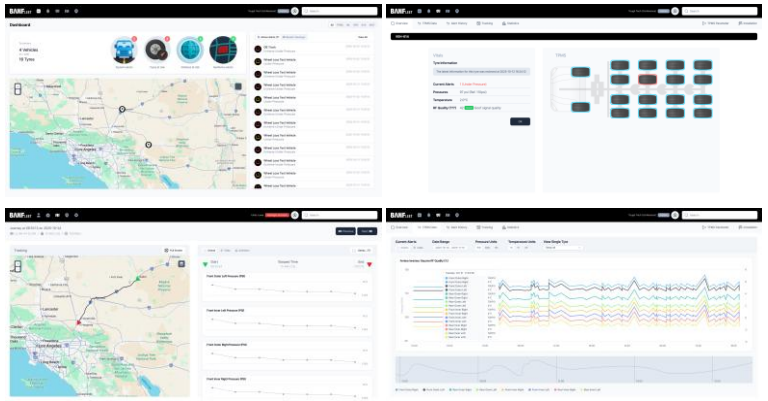
- Detect abnomal tire conditions and send instant alerts

4. Predictive Tire Maintenance

- Use tire condition trends to schedule predictive maintenance and avoid roadside failures



Fleet Overview & Alerts



BANFleet (BANF Fleet Management System)

Real Time Monitoring

- Track vehicle location and tire status overview
- Alert instantly for tire and vehicle issues

Checking Tire Condition

- Monitor detailed tire status

Journey Summary

- Review tire condition for trip
- Create and analyze reports

Why BANF

Tire Safety system

- Easy to Integration
- Global Use Cases
- Life Saving tire safety insights

